

A Quick Look at Integer Promotion in C

Data types like *char* and *short int* take a fewer number of bytes than *int*. These data types are automatically promoted to *int* or *unsigned int* when an operation is performed on them. This is known as integer promotion.



We have all heard the popular quote about the C language that, “C is an ocean, not a sea.” No matter how much of knowledge you gain in C, there’s always something new to learn about it. In this article, I focus on integer promotion in C.

What is integer promotion in C?

“Whenever any operations like arithmetic, bitwise are performed on lower rank data types such as *char* and *short*, automatically these data types will be implicitly promoted to a higher rank data type, *signed int* or *unsigned int*, until and unless explicitly typecasted.”

Let us understand the above statement with the help of the following example, Code-1:

```
1. #include <stdio.h>
2. int main()
3. {
4.     signed char x = 0xFF;
5.     unsigned char y = 0xFF;
6.     //Compare both the values of x and y
7.     if (x == y)
8.     {
9.         printf("Equal\n");
```

```
10. }
11. else
12. {
13.     printf("Unequal\n");
14. }
15. return 0;
16. }
```

Code-1: Code to compare two *char* values of different qualifiers

What will be the output of Code-1? Most people may think the answer is ‘Equal’ but, in fact, it is ‘Unequal’; that is why C is called a tricky language too. The output of Code-1 is shown in Figure 1.

Output
Code
[Satya]~/Article-IntegralPromotion/Codes gcc test 1.c [Satya]~/Article-IntegralPromotion/Codes ./a.out Unequal [Satya]~/Article-IntegralPromotion/Codes

Figure 1: Output of Code-1